Practical 9
Regression: One Numeric Predictors
Statistical Foundations for Data Science & AI

2026-10-22

Pearson Data

The question is: Is there a relationship between the heights of fathers and their sons?

Loading the Data into R

First, we'll load the data into R using the `read_excel()` function from the `readxl` package that is included in `tidyverse` in combination with the `here` package like so:

```
pacman::p_load(tidyverse,readxl)
pearson <- read_excel(here::here("data", "pearson.xlsx"))
```

Exercise

Check the data has been loaded properly by taking a look at the first few entries in the data set with the `head()` function in the chunk below.

```
head(pearson)
```

```
# A tibble: 6 × 2
  father.height son.height
          <dbl>      <dbl>
1          65.0       59.8
2          63.3       63.2
3          65.0       63.3
4          65.8       62.8
5          61.1       64.3
6          63.0       64.2
```

Visualise the Data

Before doing any further analysis, it's always good practice to start by visualising the data, to help us develop a general understanding of what to expect. So lets start by making a scatterplot.

Exercise

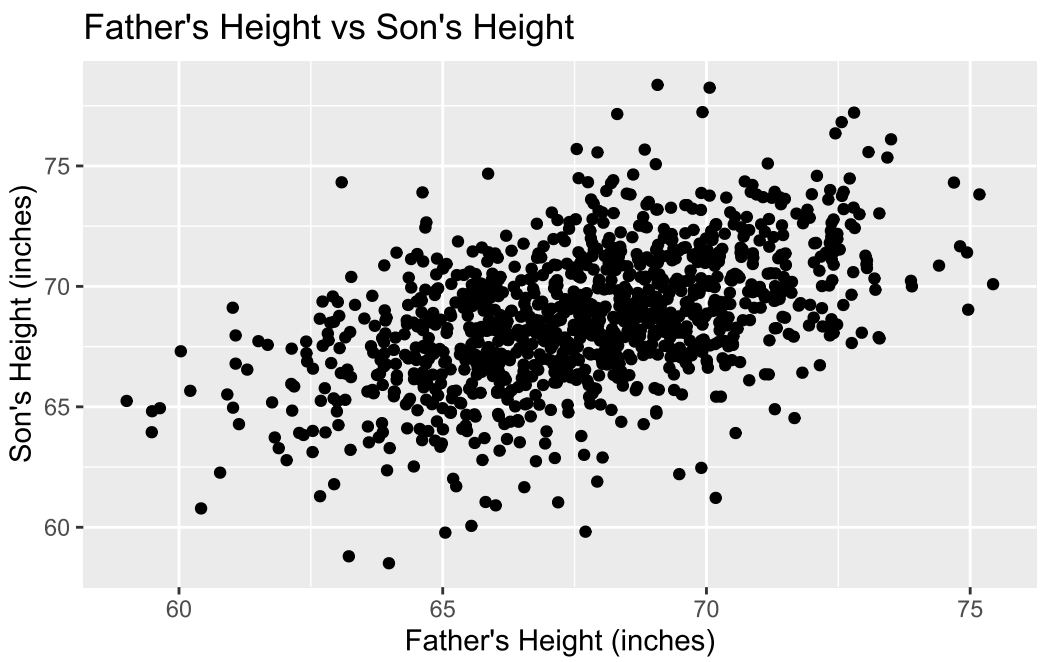
Produce a scatterplot below, with `father.height` as the explanatory or predictor variable, and `son.height` as the response.

If needed, you may copy the code for producing a scatterplot from practical 3 to use as a starting point and then edit it from there.

Describe the association between these variables, specifically including reference to the following characteristics:

- Direction(正相关/负相关): positive association. taller fathers tend to have taller sons.
- Shape: approximately linear
- Strength(相关关系强弱) moderate. 正相关趋势很明显，但点围绕线比较散开
- Outliers: a few potential outliers

```
ggplot(pearson, aes(x = father.height, y = son.height)) +
  geom_point() +
  labs(
    x = "Father's Height (inches)",
    y = "Son's Height (inches)",
    title = "Father's Height vs Son's Height"
  )
```



Fit a Linear Regression Model with One Numeric Predictor

We can fit a linear regression model to these data in R with the `lm()` command as shown in the chunk below:

用法：lm(formula,data)

formula 指定模型的自变量和因变量，<response>~<predictor>

表示希望建立指定因变量和指定自变量之间的线性关系。

```
fitlm <- lm(son.height~father.height, data=pearson)
```

> **Note**
>
> - in the chunk above we store the fitted model as a variable in R by assigning it (with the assignment operator `<-`, for which you could also use `=` if you prefer), to a variable which in this case is named `fitlm`.
> - the first argument to the `lm()` function uses the `~` operator to distinguish the response variable (to the left of the ~, in this case `son.height`) from the predictor or explanatory variable (to the right of the ~, in this case `father.height`).

We can print out a summary of the fitted regression with the `summary()` function like so:

```
summary(fitlm)
```

```

Call:
lm(formula = son.height ~ father.height, data = pearson)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.8772 -1.5144 -0.0079  1.6285  8.9685 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   33.88660    1.83235   18.49   <2e-16 ***
father.height  0.51409    0.02705   19.01   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.437 on 1076 degrees of freedom
Multiple R-squared:  0.2513,    Adjusted R-squared:  0.2506 
F-statistic: 361.2 on 1 and 1076 DF,  p-value: < 2.2e-16
```

如何解读结果

1.What are the estimates for the intercept (b_0) and slope (b_1)? Interpret these estimates in context.

b0=33.88660, b1=0.51409

斜率表示自变量每增加 1 单位，因变量的对应变化量。截距通常没有实际意义，这里也没有。

2.决定系数(R-squared)表示模型解释了因变量变异的多大比例。adjusted 是调整过后的，它考虑了模型中自变量的数量。

线性回归中 r=sqrt(R^2)

3.t value p-value

在关于斜率的假设检验中，零假设为斜率=0（即不存在线性关系），备择假设相反。

表中第一行是截距的值，第二行是斜率的。实际情况中更关注斜率，所以用第二行。

表中 p-value 显示极小，统计显著，拒绝零假设。

4.residual standard error（残差标准误）可理解为模型预测值周围的典型残差波动尺度。

Fitted Values, Residuals, and Plots

即：augment 把结果整理为数据表并附加诊断信息；

geom_point()画散点图

```
pacman::p_load(broom)

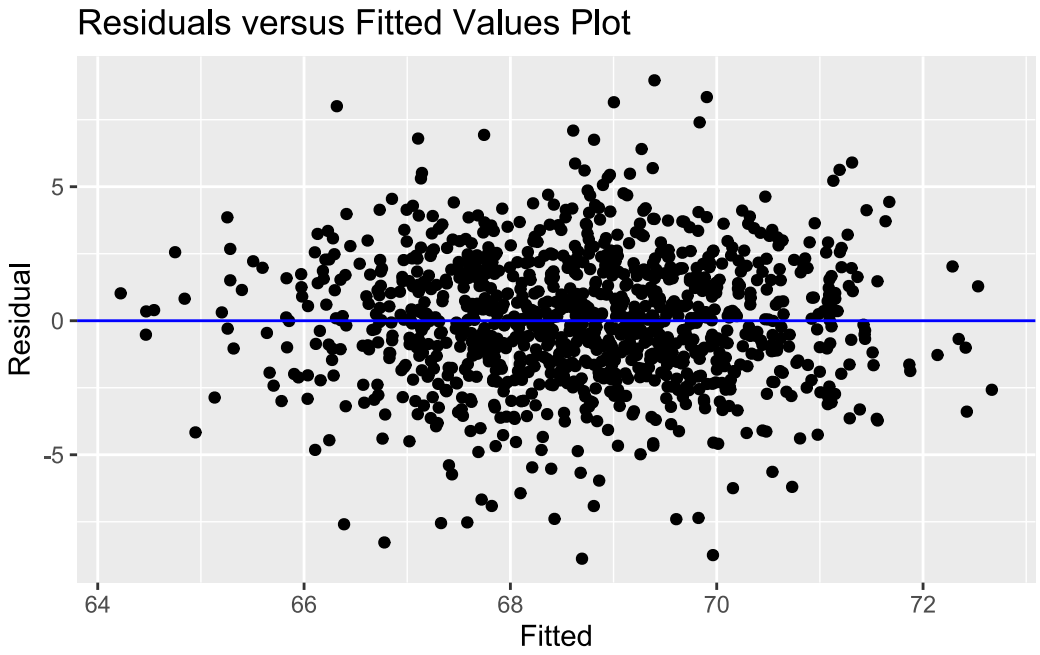
augment(fitlm) %>%
  ggplot(aes(x = .fitted, y = .resid)) +
  geom_point() +
  xlab('Fitted Value') +
  ggtitle('Residuals versus Fitted Values Plot')
```

hline 用于绘制水平参考线，指定 y 截距和颜色。

```
pacman::p_load(broom)

augment(fitlm) %>%
  ggplot(aes(x=.fitted, y=.resid)) +
  geom_point() +
  geom_hline(yintercept=0, colour ="blue") +
  ylab("Residual") +
```

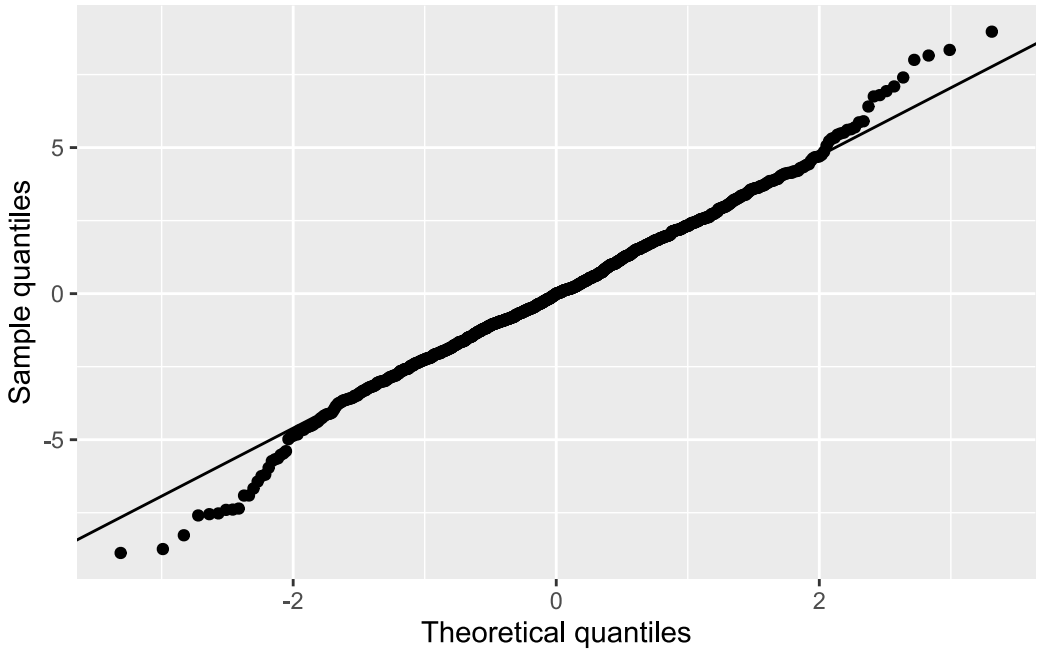
```
  xlab("Fitted") +
  ggtitle("Residuals versus Fitted Values Plot")
```



这个 residual vs fitted 图可以用来检验是否线性（散布在 0 两侧，没有明显 u 形）和方差齐性（残差的垂直范围没有持续扩大或缩小）

Q-Q plot 用于检验残差正态性。

```
augment(fitlm) %>%
  ggplot(aes(sample=.resid)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "Theoretical quantiles",y = "Sample quantiles")
```



Check Assumptions

Recall the assumptions for a linear model:

- Linearity.
- Constant variance.
- Normality.
- Independence.

We will skip the last assumption, independence, for now but in this section we will check the other three assumptions.

> Exercise
>
> Referring to the plots in the previous section, assess the three assumptions that can be assessed from these plots:
>
> - **Linearity:**
> - **Constant variance:**
> - **Normality:**

Inference for the Slope

Now that we've assessed the assumptions as best we can, we can perform a hypothesis test to check if we have evidence for a significant relationship between the height of fathers and sons based on this data.

H0 beta1=0; Ha beta1 != 0

Exercise

Perform the hypothesis test above at the 5% level of significance by entering answers where indicated by <angle-brackets> to complete the template below. Use the summary of the fitted regression model in the previous sections where appropriate to do so.

- The point estimate for <parameter> is: <point-estimate>
- The test statistic is <test-statistic> and comes from a <distribution> with <parameters>
- The p-value is <p-value>, and therefore
- We <reject/retain> the null hypothesis, since <reasoning>.
- So we conclude that <interpretation-in-context>.

Prediction

A particular father is 65 inches tall, he wants to know how tall his baby son might be when he grows up.

First need to create a new data set containing the values for which we want to make predictions, in this case we can do this with the `data.frame` function like so:

使用 dataframe 新建了一个列并保存。

```
pearson_prediction <- data.frame(father.height = 65)
```

This creates a new `data.frame` variable called `person_prediction` that has one variable, `father.height` (notice this variable is named *exactly* the same as the variable name from our previous data set — this is important), and it has one entry, the `65` we want to use to predict from.

Exercise

Print out the data set stored in the variable `pearson_prediction` in the chunk below to check it has been created correctly.

```
pearson_prediction
```

```
  father.height
1            65
```

对于所有父亲身高为 65 英寸的儿子，我们有 95% 的置信度认为他们的总体平均成年身高位于约 67.09–67.51 英寸之间。这是 CI 的计算方法。

```
predict(fitlm, newdata=pearson_prediction, interval="confidence")
```

```
       fit      lwr      upr
1 67.30265 67.09883 67.50647
```

Exercise

Using the output from the `predict()` function above, state the point estimate and 95% confidence level for the mean height of sons of fathers who are 65 inches tall based on our fitted regression model, and interpret these values in context.

SOLUTION

With 95% confidence, we believe that the mean height of the sons of fathers who are 65cm tall will be between 67.09 and 67.51 inches.

Note

Note that we round the limits of the confidence interval *outwards* in order to make our estimation of the level of confidence conservative, i.e. to increase the level of confidence and not reduce it below 95%, so the actual level of confidence for the interval we report is greater than or equal to 95% and not actually less than 95% in this case.

If we replace `interval="confidence"` with `interval="prediction"` in the `predict()` function, we can produce prediction intervals instead of confidence intervals.

Exercise

Produce a 95% prediction interval for the height of the son of a particular individual father who is 65 inches tall in the chunk below by copying the code from the example above and editing it appropriately.

Interpret the resulting prediction interval in context.

```
predict(fitlm, newdata=pearson_prediction, interval="prediction")
```

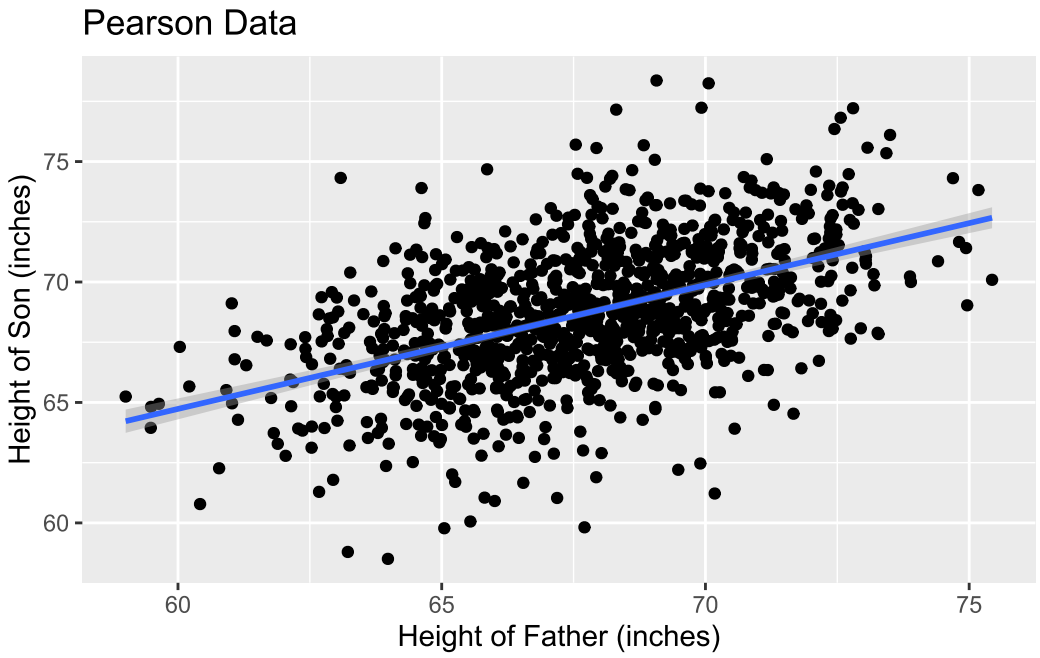
```
       fit      lwr      upr
1 67.30265 62.51737 72.08793
```

Adding a Regression Line to a Scatterplot

`geom_smooth()` 拟合曲线，lm 参数指定线性模型,level 可指定置信度。灰色区域表示置信带，相当于把每一个预测值的置信区间连续地画出来了。

```
pearson %>%
  ggplot(aes(x = father.height, y = son.height)) +
  geom_point() +
  geom_smooth(method = 'lm') +
  xlab('Height of Father (inches)') +
  ylab('Height of Son (inches)') +
  ggtitle('Pearson Data')
```

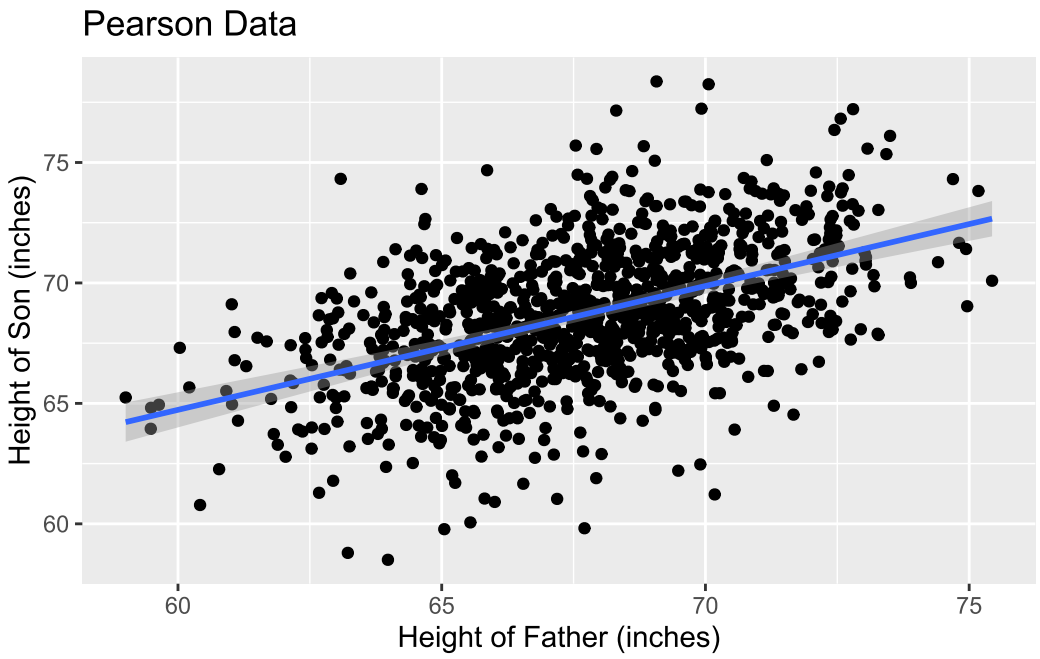
```
`geom_smooth()` using formula = 'y ~ x'
```



Notice the grey shadowed region around the blue regression line? This indicates 95% confidence intervals for the mean at different values of the explanatory or predictor variable (in this case `father.height`). We can change the level of confidence used to highlight this uncertainty by including the `level` argument to `geom_smooth()` like so:

```
pearson %>%
  ggplot(aes(x = father.height, y = son.height)) +
  geom_point() +
  geom_smooth(method = 'lm', level = 0.999) +
  xlab('Height of Father (inches)') +
  ylab('Height of Son (inches)') +
  ggtitle('Pearson Data')
```

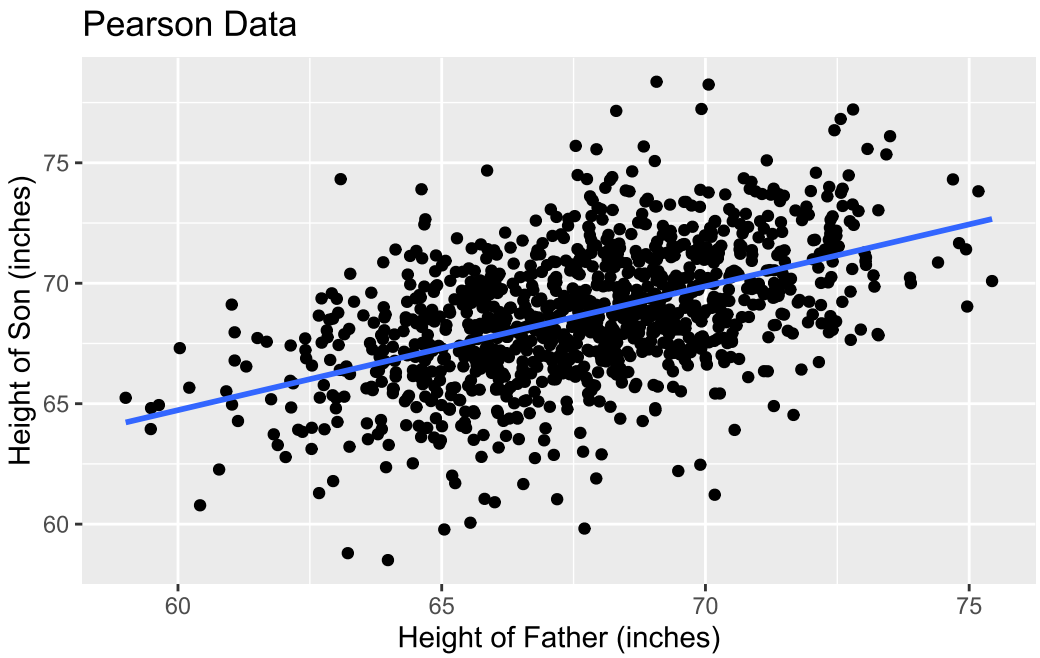
```
`geom_smooth()` using formula = 'y ~ x'
```



and we can remove it entirely if we prefer, by instead including `se = FALSE` like so:

```
pearson %>%
  ggplot(aes(x = father.height, y = son.height)) +
  geom_point() +
  geom_smooth(method = 'lm', se = FALSE) +
  xlab('Height of Father (inches)') +
  ylab('Height of Son (inches)') +
  ggtitle('Pearson Data')
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Exercise

Optional discussion point: would you expect sons to be approximately the same height as their fathers? Is this supported or contradicted by these results? Refer in particular to the estimated slope, and comment on how it relates to this question.

这玩意就是均值回归。

Diamonds Data

A statistician is looking at buying a diamond ring and wants to make sure they are getting a good price. A data set is provided with R includes details of over 50,000 diamonds that can be used to investigate the relationship between the price of diamonds (in US *dollars*) and the weight of the diamond (in *carats*).

In the previous section we used the `pearson` data set to demonstrate how to:

- fit a linear regression model,
- interpret the coefficient estimates,
- check assumptions,
- perform a hypothesis test for a coefficient, and
- make and interpret predictions.

Now its your turn. In this section, you will do all these things on the `diamond` data set. First, we'll read in the data, and take a subset of it for diamonds between 1 and 3 carets, since this is the range the statistician is considering.

```r
diamond <- ggplot2::diamonds
diamond <- diamond[diamond$carat<3,]
diamond <- diamond[diamond$carat>=1,]
head(diamond)
```

```
# A tibble: 6 × 10
  carat cut       color clarity depth table price     x     y     z
  <dbl> <ord>     <ord> <ord>   <dbl> <dbl> <int> <dbl> <dbl> <dbl>
1  1.17 Very Good J     I1       60.2    61  2774  6.83  6.9   4.13
2  1.01 Premium   F     I1       61.8    60  2781  6.39  6.36  3.94
3  1.01 Fair      E     I1       64.5    58  2788  6.29  6.21  4.03
4  1.01 Premium   H     SI2      62.7    59  2788  6.31  6.22  3.93
5  1.05 Very Good J     SI2      63.2    56  2789  6.49  6.45  4.09
6  1.05 Fair      J     SI2      65.8    59  2789  6.41  6.27  4.18
```

Visualise the Data

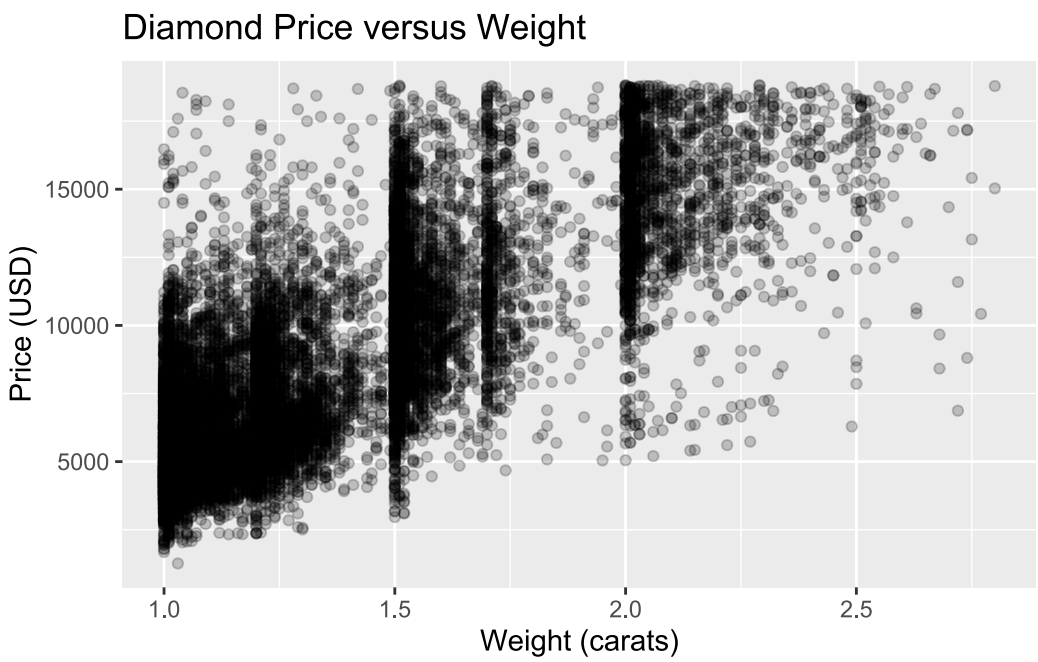
Exercise

Using `carat` as the predictor or explanatory variable, and `price` as the response variable, make a scatterplot of the `diamond` data in the chunk below.

Describe the relationship between weight and price, making specific reference to the following characteristics:

- Strength,
- Direction,
- Shape, and
- Outliers

```
diamond %>%
  ggplot(aes(x = carat, y = price)) +
  geom_point(alpha = 0.2) +
  xlab("Weight (carats)") +
  ylab("Price (USD)") +
  ggtitle("Diamond Price versus Weight")
```



Fit a Linear Regression Model

```
diamond_lm <- lm(price ~ carat, data = diamond)
summary(diamond_lm)
```

```

Call:
lm(formula = price ~ carat, data = diamond)

Residuals:
     Min       1Q   Median       3Q      Max
-13477.4  -1524.5   -477.5   1320.5  12859.5

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -3395.70      65.81   -51.6   <2e-16 ***
carat        8729.07      48.13   181.4   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2366 on 19018 degrees of freedom
Multiple R-squared:  0.6337,    Adjusted R-squared:  0.6336
F-statistic: 3.29e+04 on 1 and 19018 DF,  p-value: < 2.2e-16
```

Exercise

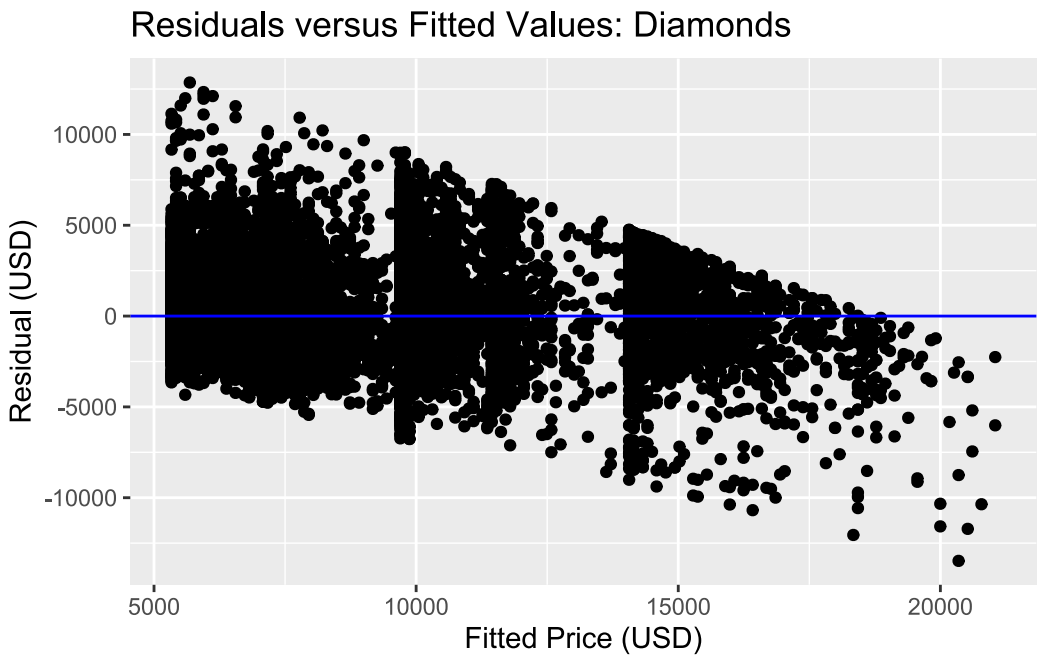
Using `carat` as the predictor or explanatory variable, and `price` as the response variable, fit a linear regression model to the `diamond` data with the `lm()` function and print a summary of the model fit with the `summary()` function. Identify the intercept and slope estimates and interpret them in context.

Check Assumptions

Exercise

Produce a fitted versus residual values plot, with the horizontal zero line highlighted in the chunk below by copying the code from the example above and editing it appropriately

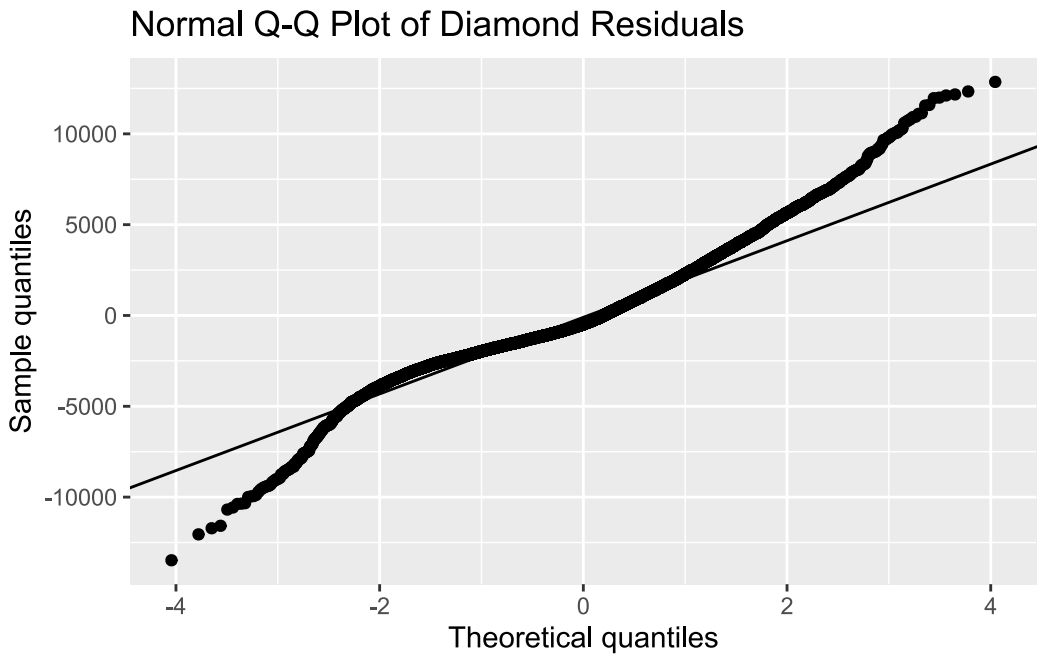
```
broom::augment(diamond_lm) %>%
  ggplot(aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, colour = "blue") +
  xlab("Fitted Price (USD)") +
  ylab("Residual (USD)") +
  ggtitle("Residuals versus Fitted Values: Diamonds")
```



Exercise

Produce a normal quantile-quantile plot of the residuals in the chunk below by copying the code from the example above and editing it appropriately

```
broom::augment(diamond_lm) %>%
  ggplot(aes(sample = .resid)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "Theoretical quantiles", y = "Sample quantiles") +
  ggtitle("Normal Q-Q Plot of Diamond Residuals")
```



Exercise

Comment on each of the four assumptions of the linear regression model:

- Linearity;
- Constant spread;
- Normality of residuals; and
- Independence.

Refer to the plots you produced above when assessing assumptions where appropriate.

Inference for the Slope

Exercise

Perform a hypothesis test at the 5% level of significance for a statistically significant linear relationship between `carat` and `price` by entering answers where indicated by <angle-brackets> to complete the template below. Use the summary of the fitted regression model in the previous sections where appropriate to do so.

- The null hypothesis is H_0 : <null-hypothesis>, and
- The alternate hypothesis is H_a : <alternate-hypothesis>,
- Where <parameters-used-in-hypothesis> is used to denote <definition-of-parameters>,
- The significance level is $\alpha =$ <significance-level>,
- The point estimate for <parameter> is: <point-estimate>
- The test statistic is <test-statistic> and comes from a <distribution> with <parameters>
- The p-value is <p-value>, and therefore
- We <reject/retain> the null hypothesis, since <reasoning>.
- So we conclude that <interpretation-in-context>.
- Since the regression assumptions were shown to be <reasonable-or-unreasonable> above, this hypothesis test may be <comment-on-validity-of-test>.

Prediction

The statistician is interested in buying a particular 2-carat diamond, and wants to know what a reasonable price would be.

Exercise

Calculate:

- a 95% confidence interval for the mean price of 2-carat diamonds, and
- a 95% prediction interval for the price of a 2-carat diamond

in the chunk below by copying the code from the examples above and editing it appropriately. Then,

- report the two intervals, and interpret them in context, and
- comment on which of the two intervals would be more appropriate to answer the statisticians question.
- the statistician has been quoted a price of $12,000 USD for the diamond, it is a fair deal?
- comment on the regression assumptions, and what impact your assessment of these assumptions has on the interpretation of these prediction intervals.

```
diamond_prediction <- data.frame(carat = 2)

# 2 克拉钻石平均价格的 95% 置信区间
predict(diamond_lm, newdata = diamond_prediction,
        interval = "confidence", level = 0.95)
```

```
       fit      lwr      upr
1 14062.45 13990.04 14134.86
```

```
# 一颗 2 克拉钻石价格的 95% 预测区间
predict(diamond_lm, newdata = diamond_prediction,
        interval = "prediction", level = 0.95)
```

```
       fit      lwr      upr
1 14062.45 9424.815 18700.08
```